

LECTURE - 1

Monday
19/8/2015

⇒ Propositions: A declarative sentence which declares a fact that could either be TRUE or FALSE.

e.g.: Lahore is capital of Pakistan ~~False~~
Islamabad is capital of Pakistan **TRUE**
 $2+1 = 3$ **TRUE**

★ Who are you? → Not proposition since it's interrogative
Do it carefully. → Command. Not proposition
He is a boy → 'He' is ambiguous.

⇒ Compound Propositions:

$3+2=5$ **and** Lahore is a city
grass is green **or** It is hot today.

CONNECTIVES	SYMBOLS	MEANING
1) Negation	$\neg$, $\sim$	NOT
2) Conjunction	$\wedge$	AND
3) Dis conjunction	$\vee$, $\oplus$	OR (inclusive), XOR (Exclusive)
4) Implication / Conditional	$\rightarrow$	if $\rightarrow$ then
5) Bi-conditional	$\leftrightarrow$	if and only if.

1) NEGATION:

P	$\neg P$
T	F
F	T

2) CONJUNCTION: AND

3) DIS CONJUNCTION: OR (inclusive) : Could be 1, the other, or both.
XOR (exclusive) : Could be either of both.

	AND	OR	XOR
$P \wedge Q$	$P \vee Q$	$P \oplus Q$	
T T	T	F	
T F	F	T	
F T	F	T	
F F	F	F	

4) CONDITIONAL: It's like a promise. 'P' made to 'Q'.
 * If P is true, Q has to be true in order for output to be TRUE.

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

* P & Q are propositions.
 'P' didn't make promise so didn't break it either so output is true.

→ Page #77 (Balk)
Uses
 $P \rightarrow Q$

- If P then Q
- P implies Q
- P is sufficient for Q
- Q if P

necessary is always Q
 • Q when P
 • Q is necessary for P
 • Q unless P
 • P only if Q

Example: $P \rightarrow Q$
 I go to the beach whenever it is a sunny day

* Even tho Q is written first but it is still $P \rightarrow Q$ (means if P is true, Q has to be true).

Inverse: $\neg P \rightarrow \neg Q$
 Converse: $Q \rightarrow P$
 Contrapositive: $\neg Q \rightarrow \neg P$

⇒ Inverse, contra., converse are always in IF THEN form.

Q: Write Inverse, converse & Contrapositive of:
 "I came to class whenever there is going to be a quiz"

TIP: first simplify in 'if P then Q' form.

~~Reverse~~
 Simplify: if "there is going to be a quiz", I came to class"

• Inverse: If there is no quiz, I won't come to class

• Converse: if I come to class, then there will be quiz

• Contrapositive: if I don't come to class, then there will be no quiz.

LECTURE - 2 | Wednesday
 19/8/25

Q: Write conditional statement, inverse, converse and contra of:

• When I stay up late, it is necessary that I sleep until noon.

Simplify: if I stay up late, then it is necessary....

Inverse: if I don't stay up late, then it is not....

Converse: if I sleep until noon, then I stayed up late.

Contrapositive: If I don't sleep until noon, then I don't stay....

Q: Truth Table of: $(P \rightarrow Q) \wedge (7P \rightarrow 7r)$

P	Q	r	$P \rightarrow Q$	$7P \rightarrow 7r$	$(P \rightarrow Q) \wedge (7P \rightarrow 7r)$
0	0	0	1	1	1
0	0	1	1	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	1	0	0

3) Biconditional: $X \leftrightarrow Y$.

Output is true if proposition 'P' and 'Q' have same truth value, else it would be False.

P	Q	$P \leftrightarrow Q$
0	0	1 (T)
0	1	0 (F)
1	0	0 (F)
1	1	1 (T)

Example: $(P \leftrightarrow Q)$

- You can take flight if and only if you buy ticket.
- $1+1=3$ iff extra is fat.
- if 2, then 3, and conversely

Q: Write Truth values: $(P \leftrightarrow Q)$

- $1+1=3$ iff $2+3=5$ False, True = False
- Milk is white iff birds lay eggs True, True = True

Q: Prop. to logical operator.

- You get an A in class, but you do not do every exercise in this Book. $P \wedge 7Q$

- You get an A in final, you do meow and you're a cat. $P \wedge Q \wedge r$

- You are cat only if you do meow or you are not a dog. $P \rightarrow (Q \vee 7r)$

$$P \rightarrow (Q \vee 7r)$$

LECTURE - 3 Monday 25/8/25.

Q: Use truth tables to prove the following:

(a) $P \rightarrow Q \equiv 7P \vee Q$.

P	Q	$P \rightarrow Q$	$7P \vee Q$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

* In order to prove expressions for which we can use Truth tables and Laws of Logic.

truth table $\rightarrow T = \text{True}$
 day / date: $F = \text{False}$
 contradiction $\rightarrow$

LAWS OF LOGICAL EQUIVALENCES:

Equivalences	Names of laws
① $p \wedge T \equiv p$ $p \vee F \equiv p$	Identity
② $p \vee T \equiv T$ $p \wedge F \equiv F$	Domination
③ $p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent
④ $\neg(\neg p) \equiv p$	Double Negation
⑤ $p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative
⑥ $p \vee (q \vee r) \equiv (p \vee q) \vee r$ $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$	Associative (Jodi Pair Change hot)
⑦ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive (Bracket kulta)
⑧ $\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's (Negation on bracket) $\neg \rightarrow \neg$ or $\neg \rightarrow \neg$ whenever
⑨ $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption should be same, std. be dif.
⑩ $p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation

day / date:

CONDITIONAL & CONDITIONAL

Equivalences	Name
⑪ $p \rightarrow q \equiv \neg p \vee q$	Conditional - Disjunction law
⑫ $p \rightarrow q \equiv \neg q \rightarrow \neg p$	Contrative
⑬ $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$	Conditional - Converse Law

Q: Simplify using laws of logic:

(a) $p \vee (\neg p \wedge q) \equiv \text{Taotology}$
 $p \vee (\neg p \vee (p \wedge q))$
 $(p \vee \neg p) \vee (p \wedge q)$
 $T \vee (p \wedge q)$
 T

Q: Verify the following:

① $\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p$
 $\neg(\neg p \vee \neg q) \wedge (p \vee q)$
 $(p \wedge q) \wedge (p \vee q)$
 $p \wedge q$

PREDICATE:

Propositional logic is not enough to express the meaning of all propositions in mathematics & natural language.

A predicate $P(x)$ is a sentence that contains a finite no. of variables and become a proposition/statement when specific values are substituted for variables.

Example:

(a) $n+1=y$

$P(n,y) : 'n+1=y' \rightarrow P(3,1) : \text{False}$

(b) $n > 5$

$P(n) : 'n > 5' \rightarrow P(2) : \text{False}$

(c) He is a great leader

$Q(n) : n \text{ is great leader}$
 $Q(\text{Imam Khen}) : \text{True/False.}$

Domain:

The domain of a predicate variable is the set of all possible values that may be substituted in place of variables.

for all, every $\forall x P(x) \rightarrow$ Universal quantifier all x True
 $\exists x P(x) \rightarrow$ Existential quantifier some x False
 $\rightarrow$ there exist, few, some

Q: What are truth values of given quantifiers:

(a) $Q(n) : 'n=n+1'$

$D = \mathbb{Z}^+$

(i) $\forall x Q(x) : \text{False}$

(ii) $\exists x Q(x) : \text{True}$

De Morgan's law of Negation / Quantifiers:

$\neg (\forall x P(x)) \equiv \exists x \neg P(x)$
 $\neg (\exists x P(x)) \equiv \forall x \neg P(x)$

Q: Negate the following.

(a) There is an honest politician

(b) All politicians are dishonest

(c) All Americans eat cheese burger.
Some Americans do not eat cheeseburger.

LECTURE-4 Wednesday 27/8/25

Q: Write negation of given statement w/out negation sign

(1) $\forall x (x^2 > x)$

$\neg (\forall x (x^2 > x)) \equiv \exists x (x^2 \leq x)$

(2) $\exists x (x^2 = 2)$

$\neg (\exists x (x^2 = 2)) \equiv \forall x (x^2 \neq 2)$

Q: Translate in English.

$C(x)$ x is comedian } domain = all people
 $F(x)$ x is funny

(a) $\forall x (C(x) \rightarrow F(x))$

Answer: All people are comedian if they are funny.
 $\rightarrow$ (a) implies $F(x)$ so: if a person is comedian, then he is funny.

(b) $\forall x (C(x) \wedge F(x))$

Every comedian is funny OR Everyone is comedian & funny.

(c) $\exists x (C(x) \rightarrow F(x))$

Some people are comedian if they are funny.

□ Quantifier over finite domain:

$\forall n \Rightarrow$ Universal: All domain values shld be true
 $P(n_1) \wedge P(n_2) \dots \wedge P(n_n)$

$\exists n \Rightarrow$ Existential: If only some parts of domain is true answer is true.

* Universal discourse means 'domain'

Q: Truth value of $\exists n P(n)$

Let $P(n) : n^2 > 10$.

Universal discourse (domain) consists of the integers not exceeding 4?

$$D = \{1, 2, 3, 4\}$$

* Since out of 1, 2, 3, 4; 4 satisfies $(4)^2 > 10$ and we've $\exists$ (existential) domain $\therefore$ **TRUE**.

$\Rightarrow$ Translating From English into logical expressions:

Q: Express the statement using predicates and quantifiers

(a) "Every student in this class has studied calculus."

* Since no domain given so we'll start from bottom.

① It is the case that every 'u' in this class has studied calculus

$C(u) : u$ studied calculus

$\rightarrow \forall u (C(u))$

② Domain: Student in this class

$S(u) : u$ is in class $\rightarrow$ shldn't be 'u'

$\rightarrow \forall u (S(u) \rightarrow C(u))$

For all students if student is in class, then he studied calculus.

(b) Some students in this class have visited Mexico.

$S(u) : u$ is a student

$M(u) : u$ have visited Mexico.

$\exists u (S(u) \wedge M(u))$

$\rightarrow$ It's not $S(u) \rightarrow M(u)$ because means answer is wrong

In translating to logical:

* $\forall u (\underline{\quad} \rightarrow \underline{\quad})$ $\rightarrow$ Bec 'for all' main samj hainay

$\exists u (\underline{\quad} \wedge \underline{\quad})$ Kuch choose krny.

$\rightarrow$ yahan already kuch options he hain to hamain choose down krnay ki zarurat nai.

(c) A student in this class knows Java, python, C++.

$J(u) : u$ knows Java

$P(u) : u$ knows Python

$C(u) : u$ knows C++

$\exists u (J(u) \wedge P(u) \wedge C(u))$

$S(u) : u$ is student in class.

Ans: $(\exists u (S(u) \wedge J(u) \wedge P(u) \wedge C(u)))$

IF $Q(x, y) : u$ knows y language.

$\exists u (S(u) \wedge Q(u, Java) \wedge Q(u, Python) \wedge Q(u, C++))$

(d) Someone is this class does not play Hockey.

$S(u) : u$ is in class

$H(u) : u$ plays Hockey

$\neg \exists u (S(u) \wedge H(u))$

(e) No one in the world can fool Fred.

$F(u, y) : u$ can fool y.

$\neg (\exists u F(u, Fred))$ OR $\forall u \neg F(u, Fred)$

LECTURE 5

day / date:

Nested Quantifiers:

Two quantifiers are said to be nested be nested if one is within the scope of other.

Note: order of quantifiers matters unless all the nested quantifiers are universal or all are existential.

NESTED	WHEN TRUE	WHEN FALSE
① $\forall x \forall y P(x,y)$	$P(x,y)$ is true for every pair x,y	There is a pair x,y for which $P(x,y)$ is False
② $\forall x \exists y P(x,y)$	For every x there is a y for which $P(x,y)$ is True	There is an x s.t. $P(x,y)$ is false for every y
③ $\exists x \forall y P(x,y)$	There is a pair x,y for which $P(x,y)$ is true	For every x , there exist some y for which $P(x,y)$ is false
④ $\exists x \exists y P(x,y)$	There is a pair x,y for which $P(x,y)$ is true	$P(x,y)$ is false for every pair x,y .

* Order matters in '2 & 3' but not in '1 and 4'.

Q: $\forall x \forall y P(x,y)$ $P(x,y) : 'x+y=0'$ $D=\mathbb{R}$

Ans: All x & y should make $P(x,y)$ TRUE for it to be TRUE.
 $P(1,2) = 1+2 \neq 0 \therefore$ **False**

Q: $\forall n \exists y P(n,y)$ $P(n,y) : 'n+y=1'$ $D=\mathbb{R}$

For all n , there should be any y for it to be true.
 $P(1,0) : 1+0=1$ **TRUE**
 $P(4,-3) : 4-3=1$

Q: $\exists x \forall y P(x,y)$ $P(x,y) : 'x+y=1'$ $D=\mathbb{R}$

For some x , all the values of y should be true.
 $P(3,5) : 3+5 \neq 1$ **FALSE**

$\exists! x$: There exist only one value of x .

day / date:

Q: Write Truth value $D=\mathbb{R}$.

(a) $\forall x \exists! y (x+y=1) \rightarrow$ TRUE
 For all x , exactly one y will satisfy.

(b) $\exists! x (x+1=0)$ TRUE

(c) $\exists! x (x>0)$ False

Q: Translate into English: Mark possibilities & construct full sentence
 Let $C(x) : x$ has a computer

$F(x,y) : x$ and y are friends
 domain: consists of all students in class

(a) $\forall x (C(x) \vee \exists y (C(y) \wedge F(x,y)))$

For every student x , x has a computer or there is a student y such that y has a computer and x & y are friends.

For every student in class has a computer or has a friend who has a computer.

(b) $\forall x \exists! y F(x,y)$

For all x , there exist only one y when s.t. x and y are friends.
 Every student have exactly one friend

(c) $\exists x \forall y \exists z ((F(x,y) \wedge F(x,z)) \wedge (y \neq z)) \rightarrow \neg F(y,z)$

There are student(s) who have only two different friends these are not friends with each other.

⇒ De Morgan's Law (Write negation)

$$Q: (a) \forall u \exists y (ay = 1)$$

$$\neg (\forall u \exists y (ay = 1))$$

$$\exists u \neg (\exists y (ay = 1))$$

$$\exists u \forall y \neg (ay = 1)$$

$$\exists u \forall y (ay \neq 1)$$

Q: Let $S(u)$: u is a student

$F(u)$: u is a faculty member

$A(u, y)$: ' u ' has asked ' y ' a question

domain(u) all people associated with your school

* differentiate variables (x and y) for student & faculty if used in same statement

Use Quantifiers to express each statement.

(a) Every student has asked Prof. Gross a question.

$$\forall u (S(u) \rightarrow A(u, \text{Prof. Gross}))$$

∴ wild've been 'u' if domain was students instead of people

(b) Every faculty member has either asked Prof. Miller

a ques or has been asked a ques by Prof. Miller

$$\forall u (F(u) \rightarrow A(u, \text{Prof. Miller}) \vee A(\text{Prof. Miller}, u))$$

$$\exists u (S(u) \wedge \neg (A(u, y) \vee A(\text{Prof. Miller}, u)))$$

(c) Some students have not asked any faculty member a ques.

$$\exists u (S(u) \wedge \forall y (F(y) \rightarrow \neg A(u, y)))$$

$$\exists F(u)$$

(d) There is a faculty member who has never asked a ques. by a student

$$\exists y (F(y) \wedge \forall u (S(u) \rightarrow \neg A(u, y)))$$

(e) Some students have asked faculty members a ques.

$$\exists u (S(u) \wedge \forall y (F(y) \rightarrow A(u, y)))$$

LECTURE - 6

Wednesday
3/9/25

□ TYPES OF INFERENCE: Already proved laws.

RULE

TABLE LOGIC

NAME

$$\textcircled{1} \frac{p \rightarrow q}{p} \therefore q$$

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$$\equiv \text{TRUE}$$

Modus Ponens
(MP)

$$\textcircled{2} \frac{p \rightarrow q}{\neg q} \therefore \neg p$$

Modus Tollens
(MT)

$$\textcircled{3} \frac{p \rightarrow q}{q \rightarrow r} \therefore p \rightarrow r$$

Hypothetical
Syllogism
(HS)

$$\textcircled{4} \frac{p \vee q}{\neg p} \therefore q$$

Disjunctive
Syllogism (DS)

$$\textcircled{5} \frac{p}{p \vee q} \therefore p \vee q$$

Addition

$$\textcircled{6} \frac{p \wedge q}{p} \therefore p$$

Simplification

RULE	Tautology	NAME
① $\frac{p}{q}$ $\therefore p \vee q$		Conjunction
② $\frac{p \vee q}{\neg p \vee r}$ $\therefore q \vee r$		Resolution

PROVE THE CONCLUSION.

Q1: "It is not sunny this afternoon and it is colder than yesterday"
"we will go swimming only if it is sunny".

"If we don't go swimming, then we will take a canoe trip"
"canoe trip", and "if we take a canoe trip then we will be home by sunset"
leads to conclusion
"we will be home by sunset". } **Conclusion**

Answer:

Variables:

p: It is sunny this afternoon
q: It is colder than yesterday
r: we will go swimming
s: we will take a canoe trip
t: we will be home by sunset.

Variable sentences in logical form.

→ Premises:

P1: $\neg p \wedge q$
P2: $r \rightarrow p$
P3: $\neg r \rightarrow s$
P4: $s \rightarrow t$

$\therefore t$ C and Logical Equivalences.

⇒ Using laws of Inference to prove the conclusion (t).

Premises	Rules Used
① $\neg r \rightarrow s$	P3
② $s \rightarrow t$	P4
③ $\neg r \rightarrow t$	HS
④ $r \vee t$	Condi. Disjunction (Lecture 3)
⑤ $r \rightarrow p$	Condi. Disjunction (Lecture 3)
⑥ $\neg r \vee p$	Resolution
⑦ $t \vee p$	P1
⑧ $\neg p \vee q$	Simplification
⑨ $\neg p$	⑦
⑩ $t \vee p$	DS
⑪ t	

Q2: Variables: "If you send me an email message, then I will finish writing a program."

"If you do not send me an email message then I will go to sleep early." and "If I go sleep early, then I will wake up feeling refreshed."

Conclusion:

"If I do not finish writing the prog, then I will wake up feeling refreshed."

→ Premises:

$$\begin{array}{l} P \rightarrow q \quad P_1 \\ \neg P \rightarrow r \quad P_2 \\ r \rightarrow s \quad P_3 \\ \hline \neg q \rightarrow s \quad C \end{array}$$

Proving:

- ① $P \rightarrow q$ P_1
- ② $\neg q \rightarrow \neg P$ Contra + of ①
- ③ $\neg P \rightarrow r$ P_2
- ④ $\neg q \rightarrow r$ HS on ② & ③
- ⑤ $r \rightarrow s$ P_3
- ⑥ $\neg q \rightarrow s$ HS ③ & ⑤

LECTURE-7 Monday 8/9/25

Q: Use rules of inference to check the validity of the argument.

'If it does not rain or if it is not foggy, then the sailing race will be held and the life saving demonstration will go on, if the sailing race is held then the trophy will be awarded' therefore 'it rained'.

⇒ Premises:

$$\begin{array}{l} P_1 \quad (\neg P \vee \neg q) \rightarrow (r \wedge s) \\ P_2 \quad r \rightarrow t \\ P_3 \quad \neg t \\ \hline C: \quad P \end{array}$$

Proving:

- ① $r \rightarrow t$ P_2
- ② $\neg t$ P_3
- ③ $\neg r$ MT on ① & ②
- ④ $\neg r \vee \neg s$ Addition on ③ : bcs we wanna go to
- ⑤ $\neg(r \wedge s)$ Demorgan's (rns) part of P1.
- ⑥ $\neg(r \wedge s) \rightarrow (r \wedge s)$ $P_1 \rightarrow$ Demorgan
- ⑦ $\neg(\neg(r \wedge s))$ MT on ⑥ & ⑦
- ⑧ $P \wedge q$ Double negation
- ⑨ P Simplification

Q: Premises (from prev Ques).

$$\begin{array}{l} \neg P \rightarrow (q \vee r) \quad P_1 \\ s \rightarrow \neg q \quad P_2 \\ s \wedge \neg r \quad P_3 \\ \hline \therefore P \quad C \end{array}$$

Proving:

- ① $s \wedge \neg r$ P_3
- ② s Simp. ①
- ③ $s \rightarrow \neg q$ P_2
- ④ $\neg q$ MP.
- ⑤ $\neg r$ Simp. ①
- ⑥ $\neg q \wedge \neg r$ Conj.
- ⑦ $\neg(q \vee r)$ Demorgan's
- ⑧ $\neg P \rightarrow (q \vee r)$ P_1
- ⑨ $\neg(\neg P)$ MT on ⑦ & ⑧
- ⑩ P Double Neg.

Rule of Inference ~~FOR~~ QUANTIFIED STATEMENTS Moving from Quantifiers & Predicates e.g. $P(x)$.

RULE	NAME
① $\forall x P(x)$ variable representing all $\therefore P(c)$ for any c constant representing any one	Universal instantiation (UI)
② $P(c)$ for any arbitrary c $\therefore \forall x P(x)$	Universal Generalization (UG)
③ $\exists x P(x)$ $\therefore P(c)$ for some c	Existential instantiation (EI)
④ $\frac{P(c)}{\therefore \exists x P(x)}$ for some c	Existential Generalization (EG)
⑤ $\forall x [P(x) \rightarrow Q(x)]$ $\therefore \frac{P(c)}{Q(c)}$	Modus Ponens (for quantified) (MP)
⑥ $\forall x (P(x) \rightarrow Q(x))$ $\therefore \frac{\neg Q(c)}{\neg P(c)}$	Modus Tollens (MT)

Q: 'Someone in this class has not read the book' and 'Everyone in this class passed first exam' implies 'Someone who passed the first exam has not read the book'.

→ Variables / Predicates

$S(x)$: x is a student in this class
 $B(x)$: x has read the Book
 $P(x)$: x has passed the first exam

⇒ Premises:

$\frac{P_1 \mid \exists x (S(x) \wedge \neg B(x))}{P_2 \mid \forall x (S(x) \rightarrow P(x))} \therefore C \mid \exists x (P(x) \wedge \neg B(x))$

always slides the values getting ANDed; then use conjunction

Proving:

① $\exists x (S(x) \wedge \neg B(x))$ P_1
 ② $S(c) \wedge \neg B(c)$ EI
 ③ $S(c)$ $Simplification (from types)$
 ④ $\forall x [S(x) \rightarrow P(x)]$ P_2
 ⑤ $P(c)$ $MP \text{ on } ③ \& ④$
 ⑥ $\neg B(c)$ $Simp. \text{ on } ②$
 ⑦ $P(c) \wedge \neg B(c)$ $Conjunction ⑤ \& ⑥$
 ⑧ $\exists x [P(x) \wedge \neg B(x)]$ EG

since 'x' and not $\rightarrow$ 'EG' & not 'UG'

Q: 'John is a student in this class' $S(x)$
 'John knows how to write program in JAVA' $J(y)$
 'Everyone who knows how to write prog. in JAVA can get a high paying job' $H(x)$
 Therefore, 'someone in this class can get a high paying job'.

⇒ Predicates

$S(n)$: n is student in class
 $J(n)$: n knows JAVA
 $H(n)$: get a high paying job.

⇒ Premises:

P_1 $S(J \wedge n)$
 P_2 $J(J \wedge n)$
 P_3 $H(n)$
 P_4 $\forall n (J(n) \rightarrow H(n))$
 C : $\exists n (S(n) \wedge J(n))$

Proving:

① $S(J \wedge n)$ P_1
 ② $J(J \wedge n)$ P_2
 ③ $\forall n (J(n) \rightarrow H(n))$ P_4
 ④ $H(n)$ MP on ② & ③
 ⑤ $S(J \wedge n) \wedge H(n)$ $conj.$ on ① & ④
 ⑥ $\exists n (S(n) \wedge H(n))$ EG on ⑤

Ex: 1.7

LECTURE-8

Wednesday
 10/9/25

INTRO TO PROOFS (types of proofs):

PROOF: A proof is a valid argument that establishes the truth of the mathematical statement.

Hypothesis of theorem

Already proved results

rule of inference

Get the truth value of the statement being proved.

TYPES

Conditional

Direct

Indirect

• Contradiction
 • Contradiction

Non-Conditional

iff

constructive

vs

uniqueness

Non-constructive

existential

Disprove by counter example.

• write statement in

$P \rightarrow Q$

Let 'P' be true;

• solve further to prove 'Q'

true as well

$\neg P \rightarrow \neg Q$

• prove from above expressions

• $P \rightarrow Q$
 $\vdash P \rightarrow Q$
 $\vdash T \rightarrow F$
 prove this. to show on $Q=F, P=F \therefore P \rightarrow Q=T$

Q: Prove that for an integer 'n', if n is odd, then n^2 is an odd integer.

Ans: [Conditional: Direct] method.

Let $n = 2k+1$, $k \in \mathbb{Z}$

$$n^2 = (2k+1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

$$\text{let } 2k^2 + 2k = r^2$$

$\therefore 2r^2 + 1$ (again a def. of odd no.).

Q: Prove that the sum of two odd integers is even

Let $m = 2k+1$, $n = 2l+1$; $k, l \in \mathbb{Z}$

odd 1

odd 2

$$m+n = (2k+1) + (2l+1)$$

$$= 2(k+l+1)$$

$$\text{let } k+l+1 = r; r \in \mathbb{Z}$$

$$\therefore 2r \text{ (def of even integer)}$$

$m+n$ is even integer

'n' is even

Q: If any integer then $(-1)^n = 1$

$$n = 2k; \quad k \in \mathbb{Z}$$

$$(-1)^n = (-1)^{2k} = 1$$

$$= \left[\frac{(-1)^2}{(1)^k} \right]^k$$

Q: Prove that the sum of two rational no. is rational.

$$m = \frac{a}{b}; \quad n = \frac{c}{d}; \quad a, b, c, d \in \mathbb{Z}; \quad b \neq 0, d \neq 0$$

$$m+n = \frac{a}{b} + \frac{c}{d} \rightarrow \frac{ad+bc}{bd}; \quad bd \neq 0$$

$$= \frac{p}{q} \quad \text{where } p = ad+bc, \quad q = bd$$

$\therefore m+n = \frac{p}{q}$ is rational.

Q: Let r and s be two rational numbers. Prove there is a number between r & s which is also rational.

$$r < s$$

$$r+n < s+n$$

$$2r < s+r$$

$$r < \frac{s+r}{2}$$

$$r < \frac{s+r}{2} < s$$

$$\text{let } n = \frac{s+r}{2}$$

$$\therefore \boxed{r < n < s}$$

Indirect

$\Rightarrow$ Contra Position (contra) $\uparrow$ Conditional Method:

Q: If n is odd integer, then n^2 is an odd integer.

Ans: Contra + : if n^2 is even integer, then n is a even integer.

* Now we'll simply use direct valid approach to prove the $79 \rightarrow 7p$ statement.

$$\text{Let } n^2 = (2k)^2$$

$$n^2 = 4k^2$$

$$n = \pm \sqrt{4k^2}$$

$$\boxed{n = \pm 2k} \quad (\text{def for even})$$

INDIRECT Conditional:

① PROOF By contra position:

• Based on $p \rightarrow q \equiv \sim 7p \rightarrow 7p$

• Assume $7p$ is true ~~and~~ & then prove $7p$ is true using direct method $7q \rightarrow 7p$

② PROOF By contradiction:

Based on Principal: Something leads to a contradiction can not be true, and if so, the opposite must be true.

① Suppose the statement to be proved is false

② Show that this suppose leads to a contradiction

③ Conclude that the statement to be proved is true.

$$p \rightarrow q$$

we need to prove $T \rightarrow T$; but if we take $p=T, q=F$, it will also make $p=F$ (contradiction occurs)

$\therefore$ we wld say p is true.

Q: Prove that if n^2 is not divisible by 25, then n is not divisible by 5.

Ans: Indirect $\rightarrow$ Contra +.

$79 \rightarrow 79$: if n is divisible by 5, then n^2 is divisible by 25

Let n is divisible by 5 ($5 \mid n$)

$$n = 5k, \quad k \in \mathbb{Z}$$

$$n^2 = 25k^2$$

$$\boxed{n^2/25 = k^2}$$

Q: Prove that if n is an integer and $3n+2$ is even, then n is even. $\rightarrow$ don't negate this

Contra +: if n is odd, then $3n+2$ is odd

Let $n = 2k+1, \quad k \in \mathbb{Z}$

$$3n+2 = 3(2k+1)+2$$

$$3n+2 = 6k+5 \rightarrow 6k+4+1$$

Let $3k+2 = r \rightarrow$ cannot be $r = 3k+5/2$ to prove it forcefully

$$\therefore 3n+2 = \boxed{2r+1}$$

def. of odd integer.

its part to change noi karyas and pr vo ulta proof hota to show

$\Rightarrow$ Using Contradiction Method.

Q: Prove if n^2 even integer, then n is an even integer

$P \rightarrow 79$: if n^2 is even integer, then n is odd.

$$n = 2k+1, \quad k \in \mathbb{Z}$$

$$n^2 = (2k+1)^2$$

$$n^2 = 4k^2 + 4k + 1$$

$$n^2 = 2(2k^2 + 2k) + 1$$

$$\therefore 2r+1 \text{ (def for odd)}$$

since n^2 got odd when we choose $n = \text{odd}$ (contradiction)

$\therefore n$ is an even integer.

Q: Let the sum of a rational no. & irrational no. is an irrational no.

Let m be rational & s be irrational

$$m = \frac{a}{b}, \quad b \neq 0, a, b \in \mathbb{Z}$$

* Opposing the 'q' part for solving with "contradiction".

Let $r+s =$ rational

$$r+s = \frac{c}{d}, \quad d \neq 0, c, d \in \mathbb{Z}$$

$$\frac{a}{b} + s = \frac{c}{d}$$

$$s = \frac{c}{d} - \frac{a}{b}$$

$$s = \frac{bc-ad}{bd}, \quad a, b, c, d \in \mathbb{Z}, b, d \neq 0$$

$\rightarrow$ Product & difference of integers is integer

$$bc-ad = p, \quad bd = q$$

$$s = \frac{p}{q} \text{ (which is rational)}$$

* since 's' is rational which is a contradiction.

to our statement of 's' being irrational

$\therefore r+s$ is irrational.

$\Rightarrow$ Proof / Disprove by Counter Example.

① For all real No. a & b , if $a < b$ then

$$a^2 < b^2$$

$$a = -5, \quad b = 2$$

$$-5 < 2$$

$$(-5)^2 < (2)^2$$

False as

$$\boxed{25 \neq 4}$$

LECTURE - 9 Monday

Ex: 1.8

PROOF METHOD & STRATEGY:

① PROOF BY CASES: Based on Rule of Inference:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$$

• It can be proved using given Tautology:

$$((p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q) \leftrightarrow ((p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q))$$

• Proof by cases cover all possibilities arise & cover all the domain

② Exhaustive Proof: Some theorems can be proved by examining a relatively small no. of examples. Such proofs are called Exhaustive proofs.

• It is special cases of proof by cases.

Q. If 'n' is an integer, then $n^2 \geq n$.* Since in 'P', you've a very large domain $\therefore$ we'll make cases:Case 1: $n \geq 1$: $(p_1 \rightarrow q)$ * Multiplying 'n' on both sides
 $n^2 \geq n$ ✓Case 2: $n = 0$: $(p_2 \rightarrow q)$ * multiply 'n' on both sides
 $n^2 = 0^2$
 $0^2 \geq 0^2$ ✓

cases entire domain

Case 3: $n \leq -1$: $(p_3 \rightarrow q)$

* Square of any -ve integer is +ve:

$$n^2 > 0 \quad n^2 > 0 > -1 \geq n$$

$$\therefore n^2 \geq n \quad \checkmark$$

Q: If 'n' is an integer, then $n^2 + 3n + 4$ is even $n \in \mathbb{Z}$

* Now we're interested in the parity (even/odd), not +ve or -ve (NBT domain split)

Case 1: n is even

$$\text{Let } n = 2k, \quad k \in \mathbb{Z}$$

$$n^2 + 3n + 4 = (2k)^2 + 3(2k) + 4$$

$$= 4k^2 + 6k + 4$$

$$= 2(2k^2 + 3k + 2)$$

$$\therefore \text{let } r = (2k^2 + 3k + 2)$$

$$\therefore 2r \leftarrow \text{even}$$

Case 2: n is odd

$$\text{Let } n = 2k + 1, \quad n \in \mathbb{Z}$$

$$n^2 + 3n + 4 = (2k + 1)^2 + 3(2k + 1) + 4$$

$$= \dots$$

$$= 2r \quad (\text{even})$$

Case 3:

$$n = 0$$

$$n^2 + 3n + 4 = 0^2 + 3(0) + 4$$

$$= 4 \quad (\text{even})$$

Q: Use a 'proof by case' to prove that $|xy| = |x||y|$

$$|a| = \begin{cases} a & a \geq 0 \\ -a & a < 0 \end{cases} \text{ Def. of modulus}$$

Case 1: let $x > 0, y > 0$

$$|xy| = xy = |x||y| \checkmark$$

Case 2: $x > 0, y < 0$

$$|xy| = -xy = x(-y) = |x||y| \checkmark \text{ By def}$$

Case 3: $x < 0, y < 0$

$$|xy| = xy = |x||y| \checkmark \text{ (as minus x minus = plus)}$$

Exhaustive Proof $\Rightarrow$ Prove that $(n+1)^3 \geq 3^n$ since domain is restricted

if n is an integer with $n \leq 4$

Case 1: $n=1 \Rightarrow (1+1)^3 \geq 3^1 \rightarrow 8 \geq 3 \checkmark$

Case 2: $n=2 \Rightarrow (2+1)^3 \geq 3^2 \rightarrow 27 \geq 9 \checkmark$

Case 3: $n=3$ just put n values.

Case 4: $n=4$ $\checkmark$

Without LOSS OF GENERALITY (WLOG):

Whenever there is a ~~symmetry~~ symmetry or arbitrary naming of variables in your theorem. By claiming 'WLOG', you can fix properties of variable & hence reduce the no. of cases.

Q: Show that for integers a & b of opposite parity then $a+b$ is odd.

* In 'proof by case' method, we would have to do $x = \text{even}, y = \text{odd}$ & then add $x = \text{odd}, y = \text{even}$

* But using WLOG, since '+' is commutative ($x+y = y+x$) $\therefore$ we can only prove one & say other is true.

Q: Show that if x & y are integers & xy and $x+y$ are both even, then x & y are both even.

$\Rightarrow$ Using Contra + would be simpler than 'if-then' form.

Contra: "if x & y are not both even then xy & $x+y$ are not both even".

x	y	Cases
odd	odd	①
odd	even	②
even	odd	②
even	even	X

Case 1: $x = \text{odd}, y = \text{odd}$

$$x = 2K+1, y = 2m+1; K, m \in \mathbb{Z}$$

$$xy = (2K+1)(2m+1)$$

$$= 4km + 2k + 2m + 1$$

$$= 2(2km + k + m) + 1$$

$$= 2r + 1 \quad (\text{odd})$$

$$n+1 = (2k+1) + 2m$$

$$= 2k+1 + 2m$$

$$= 2r+1 \quad (\text{odd})$$

Case 2: y is even, n is odd.

$$n = 2k+1, \quad y = 2m$$

$$ny = (2k+1)(2m)$$

$$= 2(2k+1)m$$

$$= 2r \quad (\text{even})$$

* Since we've already found "x+y" we no need to check again as $ny = y+x$ in both case 1 & case 2.

③ Existential Proof: A proof of a proposition of the form $\exists n P(n)$

$\Rightarrow$ 3i) Constructive proof: finding an explicit value of C for which $P(C)$ is true
 • $\exists n P(n)$ is true by $(\exists n)$ Existential Generalization.

Q. Prove that there are 100 consecutive integers that are not perfect squares.

* Let $C = 200$

$$(C)^2, (C+1)^2$$

$$= 200^2, 201^2$$

$$= 40000, 40401$$

* Since $40,401 - 40000 = 401$ $\therefore$ we can say numbers after 40001 does not have perfect squares

Q: Show that there is an integer that can be written as the sum of cubes of two integers in two different ways.
 Ans: Hit & trail.

$$2^3 + 4^3 = 8 + 64$$

$$= 72 \rightarrow \text{same integer.}$$

LECTURE -10 | Wednesday
17/9/25

Ex: 1.8

3.2

NON-CONSTRUCTIVE EXISTENCE PROOFS.

* In a non-constructive existence proof, we assume no 'c' exist which makes $P(c)$ true & derive a contradiction.

* we give argument to prove (we don't have a calculator) we do not find an element 'c' s.t. $P(c)$ is true, but rather prove $\exists n P(n)$ is true some other way.

Q: Show that there exist irrational numbers x and y such that x^y is rational.

Let $x = \sqrt{2}$ $y = \sqrt{2}$ $\Rightarrow x^y = \sqrt{2}^{\sqrt{2}}$ is rational. $\sqrt{2}$ being irrational is proved.

Case 1: If $(\sqrt{2})^{\sqrt{2}}$ is rational; we are done.

Case 2: If $(\sqrt{2})^{\sqrt{2}}$ is irrational; we will do recursion.
 * Let $x = (\sqrt{2})^{\sqrt{2}}$ & $y = \sqrt{2}$.

$$\therefore x^y = \left[(\sqrt{2})^{\sqrt{2}} \right]^{\sqrt{2}} = (\sqrt{2})^2 = 2$$

Clearly rational. $\leftarrow$ we'll stop after case 1 satisfied.

NOTE: x & y could be diff. but we should flexibly choose values that proves the qts.

Q: Show if n = rational & y = irrational;

let x : rational & y : irrational

Case 1: if $x=2$; $y=\sqrt{2}$

Case 1: if $2^{\sqrt{2}}$ is irrational; we're done

Case 2: if $2^{\sqrt{2}}$ is rational; we can plug in $2^{\sqrt{2}}$ into 'x' & can also change 'y' accordingly to prove irrational.

$x = 2^{\sqrt{2}}$, $y = \sqrt{2}/4$ ('y' is a bit bit & trail)

$$x^y = (2^{\sqrt{2}})^{\frac{\sqrt{2}}{4}} = 2^{\frac{24}{4}} = \sqrt[4]{2} \text{ irrational.}$$

Q: Show that product of two of numbers is irrational or positive. No contradiction.

$$\begin{aligned} 65^{1000} - 8^{2001} &= 3^{177} \\ 79^{1000} - 9^{2349} &+ 2^{2001} \\ 24^{4498} - 5^{8192} &+ 7^{1777} \end{aligned}$$

Ans: we've 3 numbers & 2 signs (+, -).
At least two of these 3 no. have same sign (both -ve or both +ve).

$\Rightarrow$ The product of two with same sign is Non-negative

Q: Prove or disprove that for all a, b ; if a & b are rational numbers then a^b is rational.

Counter Example:

To prove $\neg P(n)$, we take negation using Falsity of Universal Quantifier.

$\neg \forall n P(n) = \exists n \neg P(n)$ (De Morgan's law)
we find 'c' for which $P(c)$ is not true.
 $\therefore \neg P(c)$ is true.

$\rightarrow \forall a \forall b (a^b \text{ is rational})$

$$x = 2, y = 1/2 \Rightarrow 2^{1/2} = \sqrt{2} \text{ (which is irrational)}$$

$\therefore \neg$ disapprove: as $\exists n \neg P(n)$. Means that there exist some n & y where n^y is not always rational.

Existential & Uniqueness Proof:

Some theorems assert the existence of a unique solution with a particular property, $\exists! n P(n)$.

there exist only one x s.t.

$\rightarrow$ Existential: we show an element n with property exists

$\rightarrow$ Uniqueness: we show if $n \neq y$, y does not have property.

Q: Show that if $a, b \in \mathbb{R}$ & $a \neq 0$, then there is a unique real no. 'r' such that $ar + b = 0 \rightarrow$ ①

$\rightarrow$ Existential

let $r = -\frac{b}{a}$

Putting r inside ① will give $0 = 0$.

$\rightarrow$ Uniqueness:

let $r \neq s$

$ar + b = 0, as + b = 0$

$ar = as$

$r = s$ (contradiction with this)

$\Rightarrow$ SETS: $\boxed{\text{Ex } \# 2.5}$ $\downarrow$ Cardinality of $\mathbb{Z}^+$

Finite (distinct values)

Countably infinite

Uncountably infinite

• list in seq. $\downarrow$
• we can show (1 to 1) correspondence with $\mathbb{Z}^+$

• can't list them
• we cannot show 1 to 1 correspondence with $\mathbb{Z}^+$

^{count of distinct elements}

$\Rightarrow$ Cardinality: The set A & B have the same

cardinality iff there is 1-to-1 correspondence from A to B

3 element

DEF: Cardinality for infinite sets: If there is a

1-to-1 func. from A to B the

cardinality of A is less than

or equal to the cardinality of B

$\therefore |A| \leq |B|$

$\Rightarrow$ Countable Set: A set that is either

finite or has the same cardinality as the set of the integers is called

Countable $\rightarrow |S| = \aleph_0$.

• A set that is not countable is 'uncountable'.

Q: Show that the set of odd the integers is countable

• Let S be the set of odd the integers.

1) we must list elements in any order:

$S = \{1, 3, 5, 7, 9, 11, \dots\}$

2) Show 1-to-1 correspondence with the integers.

1 $\downarrow$ 3 $\downarrow$ 5 $\downarrow$ 7 $\downarrow$ 9 $\downarrow$...
+1 +2 +3 +4 +5

LECTURE - 11 Monday
29/9/25.

□ Cardinality (FINITE): No. of 'n' distinct

elements in a set

• Cardinality $|S| = n$

□ Cardinality (INFINITE): Countable infinite: S

if S has 1-to-1 correspondence with $\mathbb{Z}^+$

• Uncountable: If S does have 1-to-1 correspondence with $\mathbb{Z}^+$.

One to One: every dot in

A maps to a distinct

dot on B.

Q: Determine whether the set is finite, countably infinite or uncountable. If it is countably infinite exhibit one-to-one corresp. b/w the S and the set of the integers.

(a) Set of integers less than 100.

* Let S be the set of integers less than 100

$$S = \{99, 98, 97, 96, \dots, -3\}$$

Since the set is countably infinite as we can count; we're to show 1-1 corresp. with $\mathbb{Z}^+$.

$$S: \begin{matrix} 99 & 98 & 97 & 96 & \dots \end{matrix}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\mathbb{Z}^+: \begin{matrix} 1 & 2 & 3 & 4 & \dots \end{matrix}$$

Ans: Since we can list in sequence & show 1-1 corresp. it is countably infinite.

(b) Set of integers:

* Let S be set of integers:

$$S = \{0, 1, -1, 2, -2, 3, -3, \dots\}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\mathbb{Z}^+: \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots \end{matrix}$$

(c) Integers that are multiple of 7.

$$S = \{7, 14, 21, 28, 35, \dots\}$$

$$\mathbb{Z}^+: \begin{matrix} 1 & 2 & 3 & 4 & 5 & \dots \end{matrix}$$

Let S be set of integers which are multiple of 7.

$$S = \{-21, -14, -7, 0, 7, 14, 21, \dots\}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\mathbb{Z}: \begin{matrix} 7 & 5 & 3 & 1 & 2 & 4 & 6 & \dots \end{matrix}$$

$\therefore$ It is countable infinite

of 1, odd on left of 1.

Q: Show that the set of the rational no.s is countable:

Sol: Let S be set of the rational no.

* To show " S " as a countable infinite set, we must list its elements in a sequence:

$$r_1, r_2, r_3, r_4, \dots$$

We know by definition of rational no., " $r_i = \frac{p}{q}$ " where $p, q \in \mathbb{Z}^+$

Step 1: Form a GRID.

We take $q=1$ in first row

" " " $q=2$ in second row

" " " $q=3$ in third row

Step 2: Follow the Path (p/q) "DIAGONAL ARGUMENT"

Start listing rational $p/q=2$, followed by $p/q=3$, followed by $p/q=4$; and so on

[FIGURE]

So we don't miss any value. This is for path generation.

$$q=1: \begin{matrix} 1/1 & 2/1 & 3/1 & 4/1 & 5/1 & \dots \end{matrix}$$

$$q=2: \begin{matrix} 1/2 & 2/2 & 3/2 & 4/2 & 5/2 & \dots \end{matrix}$$

$$q=3: \begin{matrix} 1/3 & 2/3 & 3/3 & 4/3 & 5/3 & \dots \end{matrix}$$

$$q=4: \begin{matrix} 1/4 & 2/4 & 3/4 & 4/4 & 5/4 & \dots \end{matrix}$$

$$q=5: \begin{matrix} 1/5 & 2/5 & 3/5 & 4/5 & 5/5 & \dots \end{matrix}$$

STEP 3: Skip repeated terms

- For making path; we will start from $ptq=2$ & we can move in any neighbouring value (right or down) where $ptq=3$. Then we'll move further from that neighbouring value to next neighbouring value for which $ptq=4$ and so on.

- Now we'll give the same in figure we want to list in our sequence & skip the repeated terms (like: $2/1$ and $4/2$; so $4/2$ will be skipped).

STEP 4: ^{LIST 8} Show 1-1 CORRESPONDENCE with Z^+ .

LISTING IN = $1, 1/2, 2/1, 3/1, 1/3, 1/4, 2/3, \dots$

SEQ.

$Z^+ = 1, 2, 3, 4, 5, 6, 7, \dots$

Answer: Since we were able to list & make 1-1 map with Z^+ , set of the rational no. is countable infinite



Q. Show that the set of real No. is Uncountable

- we will use contradiction that set of real No. is countable to show that it is uncountable.

- Subset of every countable set is also countable.

- Take a subset ^{of $\mathbb{R}$:} $(0, 1)$ as well.

→ Now to show $S=(0, 1)$ as a countable set, we must list the elements of all real No. b/w 0 to 1.

→ let the decimal representation of real no. be:

$$r_1 = 0.d_{11} d_{12} d_{13} d_{14} \dots$$

$$r_2 = 0.d_{21} d_{22} d_{23} d_{24} \dots$$

$$r_3 = 0.d_{31} d_{32} d_{33} d_{34} \dots$$

$$r_4 = 0.d_{41} d_{42} d_{43} d_{44} \dots$$

where $d_{ij} = \{0, 1, 2, 3, \dots, 9\}$

E.g. Take

$$r_1 = 0.34157 \dots$$

$$r_2 = 0.374254 \dots$$

$$r_3 = 0.60124 \dots$$

$$r_4 = 0.006102 \dots$$

$$r_5 = 0.2826701 \dots$$

be the list of rational no's

* The 'rule' is made for digital elements of $r_1, r_2, r_3, \dots, r_5, \dots$ but we made in e.g. for prev page

day's date:

- Now by some diff decimal representation:

Rule: (we can make oneself

$$r = 0. d_1 d_2 d_3 d_4 \dots$$

form a real no.

Rule 1: $d_i = \{ d_{i-1} + 1 \}$ } could be any variation
where $d_i = \{ d_{i-1} + 1 \}$ } $\rightarrow$ operation like mod, etc.

or Rule 2: $d_i = \{ 2 \}$ } $d_{i-1} \neq 1$ } $\rightarrow$ add etc.

from r's

Number we formed: $r = 0.28268 \dots$

OR $r = 0.11211 \dots$

Ans: Since every real no. has unique decimal expansion. Therefore the real no. 'r' is not equal to any of $r_1, r_2, r_3, r_4, \dots$ Since the decimal representation differs at i^{th} place.

- No real no. $(0, 1)$ can't be listed so it's a contradiction that $(0, 1)$ is countable.

$\therefore \mathbb{R}$ is uncountably infinite.

wednesday
1/10/25

LECTURE-12

day / date:

4.1: Divisibility & Modular Arithmetic:

Def: Division: If a and b are integers with $a \neq 0$, we say 'a divides b' if there is an integer 'c' such that $b = ac$.

Q: Eg: Determine $3 \nmid 7$ and $3 \mid 12$ or not.

Theorem: Let a, b, c be integers, where $a \neq 0$,

Then

- if $a \mid b$ and $a \mid c$ then $a \mid b + c$
- if $a \mid b$, then $a \mid bc$ for all integers c .
- if $a \mid b$ and $b \mid c$, then $a \mid c$

$3 \nmid 7$: 3 does not completely divide 7
 $3 \mid 12$: 3 divides 12 completely

$\rightarrow$ If $3 \nmid 7$ $c \in \mathbb{Z}$ s.t. $7 = 3c$
 $c = 7/3$

* since $c \in \mathbb{Z}$ here $\therefore 3 \nmid 7$.

• If $3 \mid 12$ $c \in \mathbb{Z}$ s.t. $12 = 3c$
 $c = 4$

* since $c \in \mathbb{Z}$ here $\therefore 3 \mid 12$

⇒ Proof of (ii)

• let $a|b$ and $a|c$ (suppose if key based method)

By definition: as we can do $b=ad$ if $a|b$

$$b=as, \quad c=at, \quad s, t \in \mathbb{Z}$$

$$b+c = as+at$$

$$b+c = a(s+t)$$

$$\text{let } st = m, \quad m \in \mathbb{Z}$$

$$\therefore b+c = am$$

$\therefore a = b+c$ (m removed since it is a factor).

⇒ Proof of (ii)

• let $a|b$

By def.

$$b = at, \quad t \in \mathbb{Z}$$

$$bc = atc$$

$$bc = am$$

$$m = tc, \quad m \in \mathbb{Z} : m = tc$$

$$m = a|bc \quad (\text{where } m \text{ is 'c' if we see it})$$

⇒ Proof of (iii)

• let $a|b$ & $b|c$

By def: $b = as$

$$bc = asc$$

Division Algorithm:

Let 'a' be integer and 'd' be a positive integer, then there are unique integers 'q' and 'r' with $0 \leq r < d$, such that.

$$a = dq + r \quad \text{where } d \rightarrow \text{divisor}$$

$a \rightarrow \text{dividend}$

$$(a) \quad 25 \div 6 \rightarrow 25 = 4(6) + 1$$

$q = \text{quotient}$
 $r = \text{remainder}$

$$25$$

$$1(r)$$

a div d tell quotient
a mod d tell remainder

$$q = a \text{ div } d$$

$$r = a \text{ mod } d$$

Q: What are quotient & remainder

$$(a) a = 101, \quad d = 11$$

$$101 = 11(9) + 2$$

quotient

remainder

$$\text{divide } 101 / 11$$

$$\text{answer: } 9.18$$

② Take whole no. part of 9.18 i.e. = 9

$$(b) a = -11, \quad d = 3$$

$$-11 = 3(-4) + 1$$

$$3 \overline{) -11}$$

$$-12$$

$$1$$

we chose '-4' so the answer is just above the magnitude of 'a' i.e. +11

$$\text{quotient} = -4$$

$$\text{remainder} = 1$$

$$(c) \quad a = -112, \quad d = 10$$

$$-112 = 10(-12) + 8$$

$$\frac{10 \overline{) -112}}{-(-120)} + 8.$$

$$(d) \quad a = 0, \quad d = 14$$

$$0 = 14(0) + 0$$

$$(e) \quad a = 18, \quad d = 7$$

$$18 = 7(2) + 4$$

$$\frac{7 \overline{) 18}}{-14} \quad \underline{4}$$

DEF: Modular Arithmetic
If a and b are integers and m is a +ve integer then,

" a is congruent to b " if ' m divides $a-b$ '

Notation: $a \equiv b \pmod{m}$ if $m \mid a-b$.

if $18 \equiv 7 \pmod{7}$, we do $18-7$;
 18 is congruent to 7 (means: $7 \mid 18-7$)

Theorem 2:
Let a & b be integers, and let m be a +ve integer. Then $a \equiv b \pmod{m}$ iff $a \pmod{m} = b \pmod{m}$.

This means $\frac{17}{6(4)}$ should have remainder $5(a)$
day / date:

Q: Determine if:

(a) 17 is congruent to 5 mod 6.

$$6 \nmid 17-5$$

* since divisible
 $\therefore$ yes congruent

$$6 \nmid 12 = 12$$

$$\frac{117 \equiv 5 \pmod{6}}{117 \equiv 5 \pmod{6}}$$

(b) 24 is congruent to 14 mod 6

$$6 \nmid 24-14$$

* since not fully dividing $\therefore$ not congruent

$$6 \nmid 10 = 10 \cdot 6$$

Q: Determine whether the given integers are congruent to 5 mod 17

(a) 80

$$17 \nmid 80-5 = 10 \cdot 7$$

$$80 \not\equiv 5 \pmod{17}$$

(b) -29

$$17 \nmid -29-5 = 12$$

$$-29 \equiv 5 \pmod{17}$$

$$\therefore -29 \equiv 5 \pmod{17}$$

Theorem 3: Let m be +ve integer. The integers a and b are congruent mod m iff there is an integer k , s.t.:

$$a \equiv b + km \quad (\text{same thing as } a = r + dq)$$